

Whiteboard: Forwarding / Intercept design options

Options considered

1. System Level Proxy Settings

2. Network Level Interception (Routes)

3. Process Level Proxy Settings

- ~~Base URL change in settings.json~~ (ruled out)
- HTTP_PROXY & HTTPS_PROXY env vars per process (e.g. Claude in settings.json)
- Process-specific registry

4. Kernel hook / Driver

1. Minimal kernel hook code in Rust — TCP connection opened / UDP stream opened -> "socket open" event
2. Figure out the process responsible
3. From process, gather binary path & then metadata
4. IPC or local proxy to send this info to listener
5. Route the connection from step 1

Protocol sequence to handle once intercepted (client C / server S)

```
C <---- TCP/IP Connection ----> S
---- HTTP Request/Response -->
---- Upgrade to WSS ----->
<---- 100 Continue -----
<---- WSS Frame (Binary Format) ---->
```

Notes: the interception layer (option 3 or 4) needs to handle not just plain

HTTP but the HTTP->WSS upgrade handshake and subsequent binary WSS framing.